4 Group A

Paper 1: Feng et al. (2026) — Convex Language Detection

1. **Title / DOI / Year / Citations:** “Convex Low-resource Accent-Robust Language Detection in Speech Recognition”. Year: 2026. DOI: https://doi.org/10.48550/arXiv.2605.23235

```
@inproceedings{feng2026convex,
  title={Convex Low-resource Accent-Robust Language Detection in Speec
  author={Feng, Miria and Tan, William and Pilanci, Mert},
  booktitle={Forty-third International Conference on Machine Learning}
  year={2026}
}
```

2. **Method — WavLM / wav2vec2 / x-vector / ECAPA / other:** Whisper (Radford et al., 2023) and Massively Multilingual Speech (MMS). The paper proposes Convex Language Detection (CLD) implemented in JAX using ADMM.

3. **Task — accent classification / dialect ID / language variety ID:** “dialect-identification task in ASR models” and “robust language detection under dialectal variation”.

4. **Varieties close or distant? — dialects/accents of one language vs. different languages (critical):** Both. It examines “regional dialects even within high-resource languages” like Singlish (Singaporean accented English) and “5 regional dialects per language” for English and Mandarin.

5. **Reported accuracy / separability — and against what chance baseline:** “achieving 97–98% accuracy in challenging low-resource regimes”. It notes Whisper-Small accuracy drops significantly to 61.4% on Malaysian accented English despite achieving 100% on Midwestern English.

6. **Did embeddings separate close varieties well or poorly? (most important field):** Poorly (Standard Models). The source states: “current spoken dialogue systems frequently fail on under-represented dialects and accents”. Specifically, standard

models exhibit a “strong classification bias: the model defaults to predicting English, leading to 88.88% of Chinese samples being misclassified as English”.

7. **Evaluation protocol — speaker-independent (LOSO-style) or not?:** Uses an “80-10-10 train, test, validation split”.

8. **Tag:** Foundational-cite (for current ASR limitations and SSL model sensitivity).

Paper 2: Bird et al. (2019) — Accent Classification in Biometrics

1. **Title / DOI / Year / Citations:** “Accent Classification in Human Speech Biometrics for Native and Non-native English Speakers”. Year: 2019. DOI: https://doi.org/10.1145/3316782.3322780

```
@inproceedings{bird2019accent,
  title={Accent classification in human speech biometrics for native a
  author={Bird, Jordan J and Wanner, Elizabeth and Ek{\'a}rt, Anik{\'o
  booktitle={Proceedings of the 12th ACM International Conference on P
  pages={554--560},
  year={2019}
}
```

2. **Method — WavLM / wav2vec2 / x-vector / ECAPA / other:** Long Short-term Memory (LSTM) Neural Network, Random Forest, and a “DENSE NN” (complex dense neural network).

3. **Task — accent classification / dialect ID / language variety ID:** “Accent classification” and “classification of locale”.

4. **Varieties close or distant? — dialects/accents of one language vs. different languages (critical):** Close. It compares two Mexican locales (Chihuahua vs Mexico City) and two British locales (London vs West Midlands).

5. **Reported accuracy / separability — and against what chance baseline:** 94.74% accuracy for an LSTM/Random Forest ensemble. This is compared against the

"speech classification standard Hidden Markov Model (HMM)" which it outperformed by 5%.

6. **Did embeddings separate close varieties well or poorly? (most important field):** Mixed. While accuracy was high, the authors caution that models often learn "natural language patterns as a form of audible Natural Language Processing" rather than pure accent features.

7. **Evaluation protocol — speaker-independent (LOSO-style) or not?:** "10 fold cross validation". It notes that Leave-one-out (LOO) is superior but was not possible due to being "3000 times more complex" for their dataset.

8. **Tag:** Context-cite.

Paper 3: Wubet et al. (2023) — Intra-Native Accent Shared Features

1. **Title / DOI / Year / Citations:** "Intra-Native Accent Shared Features for Improving Neural Network-Based Accent Classification and Accent Similarity Evaluation". Year: 2023. DOI: https://doi.org/10.1109/ACCESS.2023.3259901

```
@article{wubet2023intra,
  title={Intra-native accent shared features for improving neural netw
  author={Wubet, Yeshanew Ale and Balram, Deepak and Lian, Kuang-Yow},
  journal={IEEE Access},
  volume={11},
  pages={32176--32186},
  year={2023},
  publisher={IEEE}
}
```

2. **Method — WavLM / wav2vec2 / x-vector / ECAPA / other:** CNN-LSTM, Bi-LSTM, GRU, and pure CNN.

3. **Task — accent classification / dialect ID / language variety ID:** "native accent identification (NAI)," "non-native English accent classification," and "accent similarity" evaluation.

4. **Varieties close or distant? — dialects/accents of one language vs. different languages (critical):** Both. It classifies 3 native English accents (USA, Canadian, UK) and 6 non-native accents (Mandarin, Italian, German, etc.).

5. **Reported accuracy / separability — and against what chance baseline:** The proposed shared-feature approach boosted accuracy by "3.7% - 7.5% on different vigorous deep learning algorithms" over the individual feature baseline.

6. **Did embeddings separate close varieties well or poorly? (most important field):** Poorly (Baseline). It highlights the "limitations of individual feature extraction methods for accent classification" and the difficulty of identifying similarities between nationalities.

7. **Evaluation protocol — speaker-independent (LOSO-style) or not?:** "Cross-validation" and "early stopping".

8. **Tag:** Context-cite (specifically for the "shared features" method to improve separability).

Paper 4: Ensslin et al. (2017) — Accent Detection in Videogames

1. **Title / DOI / Year / Citations:** "Deep Learning for Speech Accent Detection in Videogames". Year: 2017. DOI: https://doi.org/10.1609/aiide.v13i2.12968

```
@inproceedings{ensslin2017deep,
  title={Deep learning for speech accent detection in videogames},
  author={Ensslin, Astrid and Goorimoorthee, Tejasvi and Carleton, She
  booktitle={Proceedings of the AAAI Conference on Artificial Intellig
  volume={13},
  number={2},
  pages={69--74},
  year={2017}
}
```

2. **Method — WavLM / wav2vec2 / x-vector / ECAPA / other:** CNN (AlexNet) using spectrograms as input.

3. **Task — accent classification / dialect ID / language variety ID:** “detect speech accents” in NPCs within video games.

4. **Varieties close or distant? — dialects/accents of one language vs. different languages (critical):** Distant (Standard North American, British RP, and Spanish English).

5. **Reported accuracy / separability — and against what chance baseline:** “relatively low levels of accuracy (61%)” on the initial three-class set. When applied to video game files (Dragon Age: Origins), the accuracy was a mere 9%.

6. **Did embeddings separate close varieties well or poorly? (most important field):** Poorly. The results were described as having “unexpectedly poor performances,” especially when confronted with the variations of tempo and emotion found in video game voice acting.

7. **Evaluation protocol — speaker-independent (LOSO-style) or not?:** Randomly divided audio files into “75% used for training and 25% used for testing”.

8. **Tag:** Skim-only (useful for illustrating real-world/video-game deployment failures).

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4 Group B

Paper 1: Clopper et al. (2023)

1. **Title / DOI / Year / Citations:** “Second dialect acquisition and phonetic vowel reduction in the American Midwest.” Year: 2023. DOI: https://doi.org/10.1016/j.wocn.2023.101243

```
@article{clopper2023second,
  title={Second dialect acquisition and phonetic vowel reduction in th
  author={Clopper, Cynthia G and Burdin, Rachel Steindel and Turnbull,
  journal={Journal of Phonetics},
  volume={99},
  pages={101243},
  year={2023},
```

```
  publisher={Elsevier}
}
```

2. **Which measure — vowel formants / VOT / rhythm metrics / Pillai / sibilant moments / distance-matrix-dendrogram:** Vowel duration and first and second formant frequencies (F1, F2).

3. **Normalization used — Lobanov or other?:** Lobanov normalization. The source states: “F1 and F2 estimates were normalized by talker using z-scores (Lobanov, 1971)”.

4. **How the distance/clustering was built:** k-means clustering using Euclidean distance in a six-dimensional space. The source specifies: “The distance metric used in the k-means clustering analysis was Euclidean distance in a six-dimensional space, with one dimension for each of the two formants for each of the three vowels”.

5. **Principal findings:** Confirmed the Northern Cities Shift among Northern talkers. It found that 60% of Northern transplants to the Midland were assigned to the non-NCS cluster, suggesting second dialect acquisition.

6. **Foundational vs. applied:** Application (precedent for k-means clustering of normalized vowel spaces).

7. **Tag:** Context-cite.

Paper 2: Tanner et al. (2022)

1. **Title / DOI / Year / Citations:** “Multidimensional acoustic variation in vowels across English dialects.” Year: 2022. DOI: https://doi.org/10.18653/v1/2022.sigmorphon-1.8

```
@inproceedings{tanner2022multidimensional,
  title={Multidimensional acoustic variation in vowels across English
  author={Tanner, James and Sonderegger, Morgan and Stuart-Smith, Jane
  booktitle={Proceedings of the 19th SIGMORPHON workshop on computatio
  pages={72--82},
  year={2022}
```

```
}
```

2. **Which measure:** Vowel formants (onset/offset), duration, and trajectory shape measures (Vector Length, Offset, Angle).

3. **Normalization used — Lobanov or other?:** Lobanov normalization. The source states: “The remaining middle 60% of the vowel (13 points) was then z-score normalized against all vowels produced by the speaker (‘Lobanov normalization’, Lobanov, 1971)”.

4. **How the distance/clustering was built:** Support Vector Machines (SVMs) trained on combinations of measurements to classify 21 dialects.

5. **Principal findings:** Accuracy is improved when both formant position and trajectory shape are used. Duration provides a significant increase in accuracy across all vowels when used alongside formant information.

6. **Foundational vs. applied:** Application (precedent for including dynamic trajectory information in dialect classification).

7. **Tag:** Context-cite.

Paper 3: Sciarretta (2024)

1. **Title / DOI / Year / Citations:** “Dialectometry-based classification of the Central–Southern Italian dialects.” Year: 2024. DOI: https://doi.org/10.1017/jlg.2024.7

```
@article{sciarretta2024dialectometry,
  title={Dialectometry-based classification of the Central--Southern I
  author={Sciarretta, Antonio},
  journal={Journal of Linguistic Geography},
  volume={12},
  number={1},
  pages={13--23},
  year={2024},
  publisher={Cambridge University Press}
}
```

2. **Which measure:** Numerical evaluation of 18 phonetic features using Manhattan distance.

3. **Normalization used — Lobanov or other?:** Categorical encoding (digits 0 to P-1) with a vector of weights.

4. **How the distance/clustering was built:** k-means algorithm applied to 647 “homogeneous areas” in an 18-dimensional space. Silhouette analysis was used to select the number of clusters (K=8).

5. **Principal findings:** Resulted in a nine-group classification that is “less subjective, more precise, and more comprehensive than traditional ones based on selected isoglosses”.

6. **Foundational vs. applied:** Application (precedent for k-means dialectometry on manually identified phonetic traits).

7. **Tag:** Context-cite.

Paper 4: Sung et al. (2025)

1. **Title / DOI / Year / Citations:** “Applying the state-of-the-art tonal distance metrics to a large dialectal dataset.” Year: 2025. DOI: Not available — no DOI is listed for this book chapter (Language Science Press); ISBN/publisher record only.

```
@article{sungapplying,
  title={Applying the state-of-the-art tonal distance metrics to large
  author={Sung, Matthew and Prokic, Jelena and Chen, Yiya}
}
```

2. **Which measure:** Tonal distance metrics (including Onset-Contour-Offset (OCO) and Gandour-Harshman-Tang (GH-T)).

3. **Normalization used — Lobanov or other?:** Normalized Manhattan distance and Levenshtein distance.

> Exact line from PDF: “all the differences are divided by 10 so that the range of differences are kept between 0 and 1, just like the other metrics”.

4. **How the distance/clustering was built:** Ward's method for cluster analysis.

> Exact line from PDF: "cluster analysis (Ward's method) was performed on the segmental classification as well as the classification for each tone distance measure".

5. **Principal findings:** Found that OCO and GH-T are meaningful but require improvement; noted that tones do not follow the same patterns as segments.

> Exact line from PDF: "Our results show that the OCO and GH-T representations make meaningful contributions to dialectometry, but they are not yet ready for the use in measuring tonal distances."

6. **Foundational vs. applied:** Application (precedent for systematic comparison of tonal distance metrics in dialectometry).

7. **Tag:** Context-cite.

Paper 5: Cheng et al. (2023)

1. **Title / DOI / Year / Citations:** "A comparative study of English vowel shifts and vowel space area among Korean Americans in three dialect regions." Year: 2023. DOI: https://doi.org/10.1017/jlg.2023.1

```
@article{cheng2023comparative,
  title={A comparative study of English vowel shifts and vowel space a
  author={Cheng, Andrew and Jeon, Lisa and Kim, Dot-Eum},
  journal={Journal of Linguistic Geography},
  volume={11},
  number={1},
  pages={38--52},
  year={2023},
  publisher={Cambridge University Press}
}
```

2. **Which measure:** Vowel Space Area (VSA) and Euclidean distance between short front vowels and an anchor vowel (FLEECE).

3. **Normalization used — Lobanov or other?:** Lobanov normalization. The source states: “normalized all the formant measurements using Lobanov (Z-score) normalization (Lobanov, 1971)”.

4. **How the distance/clustering was built:** Calculated VSA using Heron’s formula for a five-corner vowel polygon.

5. **Principal findings:** Californian speakers had the smallest vowel spaces, while Georgian speakers had the largest. Georgian speakers’ VSA patterned along axes of social network and ethnic orientation.

6. **Foundational vs. applied:** Application (precedent for VSA/Heron’s formula analysis in regional identity).

7. **Tag:** Context-cite.

Paper 6: Shim & Nerbonne (2021)

1. **Title / DOI / Year / Citations:** “dialectR: Doing Dialectometry in R.” Year: 2021 (published 2022). DOI: https://doi.org/10.18653/v1/2022.vardial-1.3

```
@inproceedings{shim2022dialectr,
  title={dialectR: Doing dialectometry in R},
  author={Shim, Ryan Soh-Eun and Nerbonne, John},
  booktitle={Proceedings of the Ninth Workshop on NLP for Similar Lang
  pages={20--27},
  year={2022}
}
```

2. **Which measure:** Edit distance (Levenshtein) of transcriptions and MFCC-based acoustic distance.

3. **Normalization used — Lobanov or other?:** Length-normalized edit distance and cepstral means and variance normalization for MFCCs.

4. **How the distance/clustering was built:** Multidimensional scaling (MDS) and hierarchical clustering implemented in an open-source R pipeline.

5. **Principal findings:** Demonstrated that MFCC-based acoustic distance correlates strongly (r = −0.71) with human judgments of native-likeness.

6. **Foundational vs. applied:** Method-defining (provides the toolkit for contemporary R-based dialectometry).

7. **Tag:** Foundational-cite.

Paper 7: Bompolas & Melissaropoulou (2025)

1. **Title / DOI / Year / Citations:** “Understanding Dialectal Variation in Contact Scenarios Through Dialectometry.” Year: 2025. DOI: https://doi.org/10.3390/languages10010013

```
@article{bompolas2025understanding,
  title={Understanding dialectal variation in contact scenarios throug
  author={Bompolas, Stavros and Melissaropoulou, Dimitra},
  journal={Languages},
  volume={10},
  number={1},
  pages={13},
  year={2025},
  publisher={MDPI}
}
```

2. **Which measure:** Relative distance value (RDV) for grammatical features and PMI Levenshtein distance for lexical features.

3. **Normalization used — Lobanov or other?:** Standardized features to have a mean of zero and standard deviation of one.

4. **How the distance/clustering was built:** Multiple regression on distance matrices (MRM) evaluating geography, population size, and variety type.

5. **Principal findings:** Geography is the strongest driver for lexical divergence. For grammatical distances, the Turkish population in the province is the most significant predictor.

6. **Foundational vs. applied:** Application (precedent for using MRM to separate geographic from macro-social factors in dialectometry).

7. **Tag:** Context-cite.
